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Mitosis & Meiosis

The Cell Cycle

Cells reproduce by a process called **cell division**.the **cell cycle** is the sequence of stages of growth and division that a cell undergoes.the three stages of the cell cycle include **interphase**,**mitosis** and **cytokinesis**.

Interphase is the first stage of the cell cycle and the period before cell division.during this phase,the cell matures,copies its dna,and prepares to divide.

Prophase
the chromatin condenses and spindle fibers form at each side of the cell,the nuclear membrane breaks apart.

Centromere
Chromatin
Chromatids

Chromatin

The Cell Cycle

Centriole

Interphase
Cell Growth
DNA Replication

Metaphase
The chromosomes line up along the center of the cell and the spindle fibers attach to each chromosome at the centromere.

Daughter Cells

Cytokinesis
during this stage the cell membrane pinches in at the middle of the cell dividing the cell into two separate daughter cells.each daughter cell gets half of the cell organelles and an identical set of chromosomes.

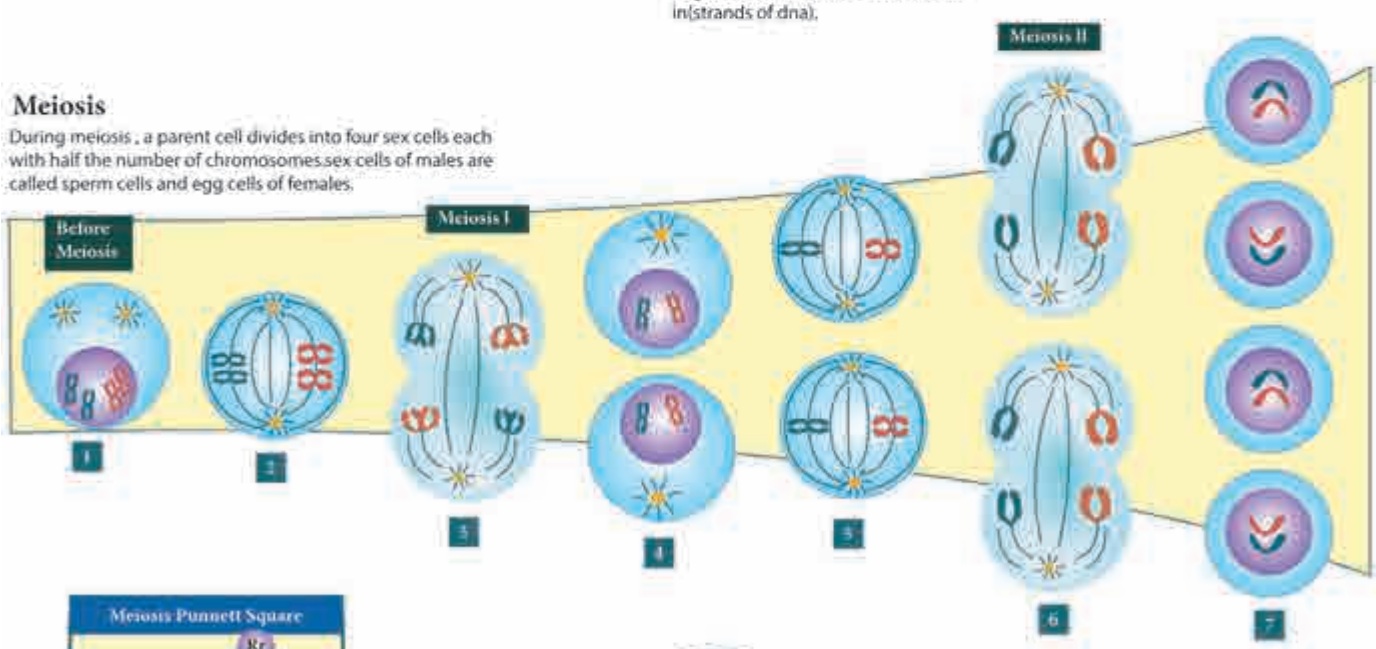
Cytokinesis
Preparation For Mitosis
Mitosis

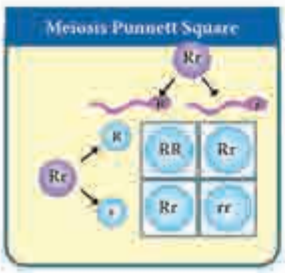
Anaphase
the spindle fiber splits the centromere and the chromatids move to opposite sides of the cell.

Telophase
the chromosomes loosen on each side and the nuclear membrane begins to form around the chromatin(strands of dna).

Meiosis

During meiosis , a parent cell divides into four sex cells each with half the number of chromosomes.sex cells of males are called sperm cells and egg cells of females.







Mitotic cell
A cell preparing to undergo division, Notice the dense chromosome